SMART INDIA HACKATHON 2026

Problem Statement ID: **SIH26263**

Problem Statement Title: Development of an APAAR ID-Based Student Data Management and Verification Platform for AICTE-Approved Institutions

Theme: Smart Automation

PS Category: Software

Team ID: TCE_Yquanto

Team Name: **TCE_Yquanto (Thiagarajar College of Engineering, Madurai)**

Source Repository: **https://github.com/naveeshpcbin/APAAR-ID-BASED-SYSTEM**

Architecture Baseline: Next.js 14 + FastAPI + PostgreSQL (20 Tables, RLS) + Deterministic 3-Way Reconciliation

Verification Status: **64/64 Passing Automated Pytest Suite across 14 Verification Gates**



TCE_Yquanto

PROPOSED SOLUTION: THE PRAMAAN-AICTE RECONCILIATION SYSTEM

SMART INDIA HACKATHON 2026

1. REGULATORY CHALLENGE ADDRESSED

- **Fragmented Institutional Data:** Divergent college ERP spreadsheet schemas, non-standard column headers, and unverified declarations.
- **Reconciliation Bottlenecks:** Difficulties cross-verifying declared enrolments against actual student records and AICTE approval ceilings.
- **Intake & Duplicate Risks:** Lack of automated visibility into over-intake breaches, quota diversion, and concurrent multi-college claims.

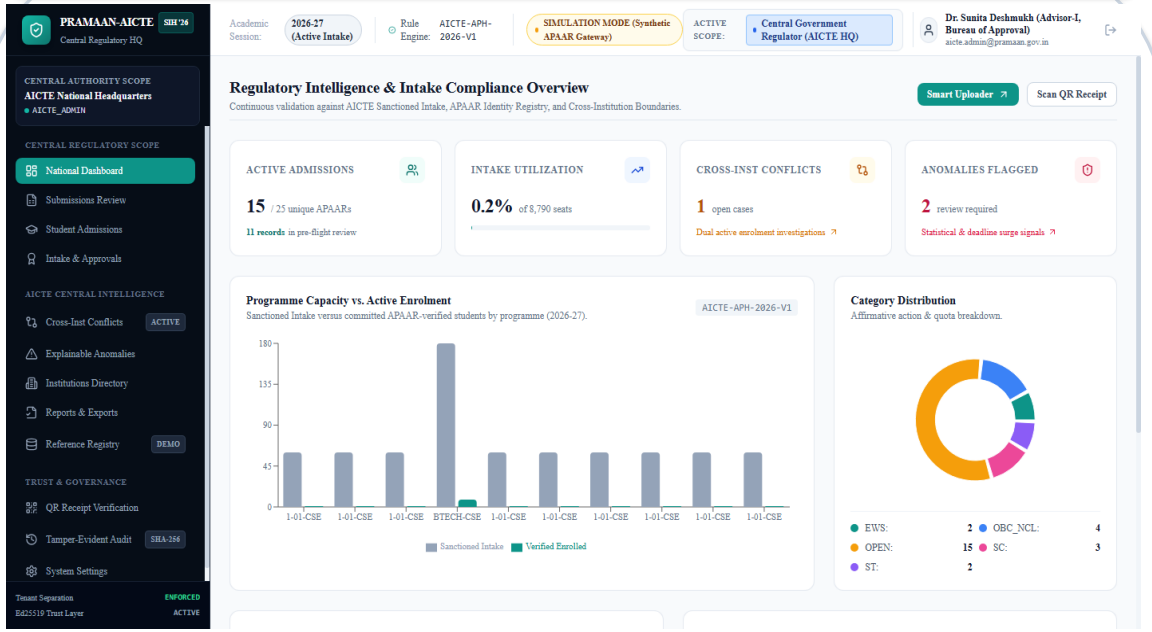
2. WHAT PRAMAAN-AICTE IMPLEMENTS

- **Decoupled Institutional Portals:** Role-separated desks for College submission and Central AICTE regulatory governance.
- **Automated Ingestion Desk:** CSV/Excel file parsing with resilient column alias mapping into standard staging rows.
- **Deterministic 3-Way Reconciliation:** Pure Python comparison engine cross-checking declarations, staged student records, and approved capacity.
- **Persisted Validation Runs:** Stores row-level findings with precise classification: approved, quota breach, syntax error, or duplicate.

3. KEY WORKFLOW & GOVERNANCE VALUE

- **Explainable Row-Level Findings:** Direct, transparent feedback explaining specific discrepancy causes without opaque batch rejection.
- **Quarantine & Resubmission:** Interactive in-browser quarantine sandbox allowing institutions to correct data before council commit.
- **Controlled Central Decisions:** Council review workflow supporting formal decisions: approve, return for correction, or reject.
- **Audit Traceability:** Complete validation run history and append-only sequential SHA-256 audit chaining across actions.

AICTE CENTRAL APEX COMMAND & RECONCILIATION DASHBOARD



✓ **Central Intake Intelligence:** Live monitoring of sanctioned intake vs enrolled students across B.Tech, MCA, and Diploma.

✓ **Deterministic Review Desk:** Instant identification of seat limit breaches, unverified candidates, and duplicate claims.

✓ **Multi-Tenant Governance:** Seamless institutional isolation between AICTE Central Regulators, DTU Delhi, and COEP Pune.

✓ Deterministic Python Core

⚖️ 3-Way Cross-Reconciliation

🔒 Cryptographic Verification & Audit

🛡️ 64/64 Verified Passing Tests

TECHNICAL APPROACH AND SYSTEM ARCHITECTURE

5-STAGE RECONCILIATION & GOVERNANCE PIPELINE

STAGE 1: INSTITUTIONAL INGESTION & MAPPING

Institutional user authenticates via JWT; uploads CSV/Excel student admissions file. Header normalizer maps diverse column aliases into standard schema.

STAGE 2: DETERMINISTIC PREFLIGHT SCREENING

Fast preflight validation checks mandatory identifier syntax, field types, and isolates corrupted records into quarantine staging.

STAGE 3: 3-WAY RECONCILIATION ENGINE

Deterministic Python engine compares staged records against institutional declarations, approved programme capacities, and synthetic reference records.

STAGE 4: FINDINGS PERSISTENCE & SANDBOX

Persists validation run records, row-level error findings, and enables in-browser quarantine sandbox for institutional dispute resolution.

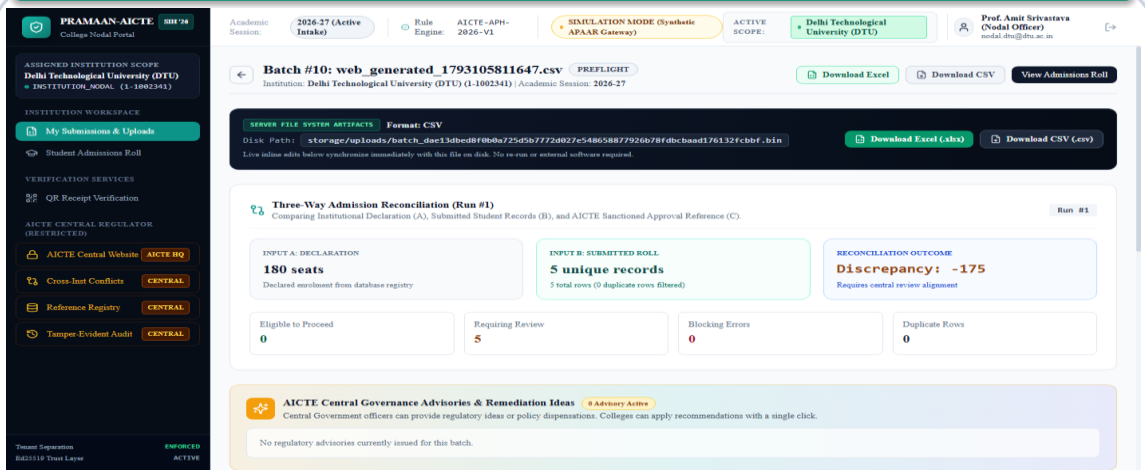
STAGE 5: CENTRAL REVIEW & VERIFICATION PROOFS

AICTE Central reviews batch: approves, returns for correction, or rejects. Generates Ed25519-signed verification receipts and appends SHA-256 audit events.

VERIFIED 4-LAYER SYSTEM ARCHITECTURE

- **Presentation Layer:** Next.js 14 (App Router), TypeScript, Tailwind CSS, Recharts for analytics.
- **Application & API:** FastAPI (Modular Monolith), Python 3.10+, Pydantic v2, Uvicorn, 73 endpoints.
- **Persistence & Schemas:** SQLAlchemy ORM, PostgreSQL (20 tables with RLS enabled) / SQLite WAL mode.
- **Cryptography & Security:** Ed25519 (RFC 8032 signing), Keyed HMAC-SHA256, sequential SHA-256 audit chaining.

INTERACTIVE QUARANTINE SANDBOX & ROW-LEVEL FINDINGS



Genuine Implemented UI: In-browser editable spreadsheet sandbox isolates validation findings before central commit.

FEASIBILITY, VIABILITY AND RISK MITIGATION

IMPLEMENTATION FEASIBILITY & VERIFIED MATURITY

- **Architectural Feasibility:** Zero-install web architecture. Standard modern browser access across Windows, Linux, and macOS. Decoupled Next.js client and FastAPI modular backend.
- **Verified Test Suite (64/64 Passing):** Exhaustive automated verification across 14 test suites covering authentication, institution onboarding, batch upload, deterministic reconciliation, security, and reference data.
- **Deterministic Logic Integrity:** Reconciliation logic is pure, explainable Python with explicit rule sets for sanctioned intake and supernumerary quotas; avoids opaque or non-deterministic AI decisions.
- **Synthetic Data Testing Engine:** Database-backed synthetic APAAR identities and editable approval/declaration models enable comprehensive pre-deployment stress testing and repeatable verification.
- **Dual-Dialect Database Support:** SQLAlchemy ORM architecture with dynamic driver fallback (psycopg3/psycopg2) verified on local SQLite WAL mode and configured PostgreSQL schemas (20 tables, RLS active).
- **Transparent Deployment Status:** Application codebase, schemas, and test suites 100% verified locally; cloud Render web deployment is currently pending resolution of cloud environment DNS configuration.

GOVERNANCE RISKS & TECHNICAL COUNTERMEASURES

⚠️ **Risk: Inconsistent Institutional Spreadsheets:** Colleges use varying ERP schemas and column headers (e.g. apaar_id vs uid vs student_apaar).

→ **Control:** Header alias normalizer maps 50+ divergent column variants into standard canonical schema automatically.

⚠️ **Risk: Student Identifier Privacy Exposure:** Storing plaintext 12-digit student APAAR identifiers risks data leakage under DPDP Act 2023.

→ **Control:** Keyed HMAC-SHA256 pseudonymized indexing and dynamic masking (XXXX-XXXX-9012) ensure zero plaintext PII exposure.

⚠️ **Risk: Multi-Institution Concurrent Admissions:** Students admitted concurrently across different institutions leading to seat blocking.

→ **Control:** Cross-institution collision radar cross-checks active admissions against centralized reference records and flags duplicates.

⚠️ **Risk: Regulatory Non-Repudiation & Tampering:** Unauthorized insider tampering with approved intake capacity or admission records.

→ **Control:** Append-only sequential SHA-256 hash chaining (Hash_n = SHA256(Record_n || Hash_n-1)) mathematically exposes any tampering.

⚠️ **Risk: Database Driver & Cloud Hosting Mismatch:** Container deployment failures caused by PostgreSQL driver compatibility or DNS misconfiguration.

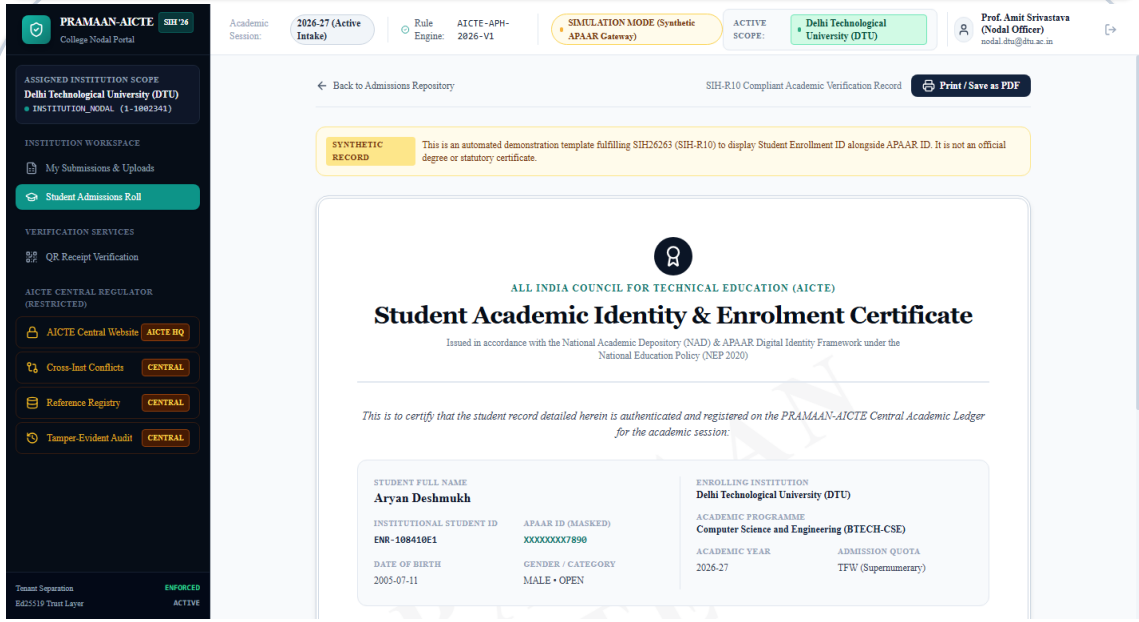
→ **Control:** Implemented psycopg3/psycopg2 dynamic driver fallback with automated fallback logging and sanitized connection strings.

MULTI-STAKEHOLDER IMPACT & DEGREE VERIFICATION

MULTI-STAKEHOLDER REGULATORY IMPACT

- **AICTE & Regulatory Councils:** Comprehensive real-time visibility into institutional enrolment versus sanctioned intake quotas; early detection of unauthorized excess enrolments and quota breaches.
- **Approved Institutions:** Standardized data submission, instant row-level discrepancy explanations, and controlled resubmission workflows replacing prolonged paper audit cycles.
- **Students & Deserving Candidates:** Transparent tracking protects legitimate seat allocations and safeguards affirmative action quotas from unauthorized commercial reallocation.
- **Academic Verifiers & Employers:** Fast offline verification via QR code and cryptographically signed verification receipts without requiring manual institutional registrar confirmation.
- **Social Equity & Affirmative Action:** Rigorous enforcement of statutory and supernumerary quotas (TFW, EWS, Kashmiri Migrants) against arbitrary substitution.
- **Administrative & Paperless Efficiency:** 100% digital data reconciliation replaces manual paperwork, physical affidavits, and physical institutional auditing logistics.

SIH REQUIREMENT 10: DEGREE CERTIFICATE VERIFICATION



✓ **Dual Identity Display:** Certificates present Student Enrolment Roll No (ENR-108410E1) alongside masked APAAR ID (XXXXXXXX7890) and category quota status.

✓ **Asymmetric Ed25519 QR Code:** Embedded high-density QR code carries a digitally signed verification token; verifiers validate data integrity against AICTE public key offline.

✓ **Integrity Boundary Clarification:** Digital signature confirms platform-level record integrity and non-tampering; explicitly distinguished from official government identity confirmation.

RESEARCH, REFERENCES AND PROJECT RESOURCES

AUTHORITATIVE STANDARDS & REGULATORY CITATIONS

- **AICTE Approval Process Handbook (APH 2026-27)**
  Statutory norms for approved institutional intake capacity, supernumerary seat limits (TFW, EWS, Kashmiri Migrants), and programme approval guidelines.
- **National Education Policy (NEP 2020) & APAAR / ABC**
  Government policy framework establishing the 12-digit Automated Permanent Academic Account Registry as the lifelong national academic identifier.
- **Digital Personal Data Protection Act (DPDP Act 2023)**
  Data fiduciary compliance, purpose limitation, identifier pseudonymization, dynamic masking, and student consent management frameworks.
- **IETF RFC 8032: Ed25519 Cryptography**
  Specification for Edwards-Curve Digital Signature Algorithm (Curve25519) providing mathematical non-repudiation and instant receipt verification.
- **NIST SP 800-107: Recommendation for Secure Hashing**
  Cryptographic guidelines for sequential SHA-256 hash chaining utilized in the tamper-evident regulatory audit trail.

SCOPE DIFFERENTIATION (PRIOR ART VS PRAMAAN-AICTE)

- **Validation Scope:** Generic uploads check syntax only. PRAMAAN performs 3-way reconciliation (Declarations × Uploads × Sanctioned Capacity).
- **Reference Data:** Traditional workflows rely on static assumptions. PRAMAAN integrates dynamic database references with revalidation.
- **Finding Resolution:** Standard systems use opaque batch rejections. PRAMAAN provides row-level explainable findings and an editable quarantine sandbox.
- **Verification Output:** Conventional portals yield unverified spreadsheets. PRAMAAN produces Ed25519-signed verification receipts and QR proofs.

VERIFIED PROJECT RESOURCES & ENVIRONMENT STATUS

✓ **Open-Source GitHub Repository:** https://github.com/naveeshpcbin/APAAR-ID-BASED-SYSTEM (Complete source code, 20 application models, Alembic migrations, test suites)

✓ **Interactive Backend API Docs:** FastAPI OpenAPI 3.1 documentation available at `/docs` (73 endpoints registered across 67 paths).

✓ **Automated Test Verification:** 64/64 pytest automated tests passing locally across 14 test suites covering all core workflows.

✓ **System Integration Status:** • External APAAR / DigiLocker / ABC API: SIMULATED (Synthetic reference dataset).

• Cloud Render Deployment: PENDING RESOLUTION (Host DNS configuration in Render container).